> We could use AI to help us provide better help to operators with network resources.
>
> —***Luis Jorge Romero,*** *director-general, ETSI*

AI based adaptive networks can help in fibre capacity analysis and optimisation. Policy based matching of channel and wavelength capacity improves the efficiency and adaptive planning of optical networks. Providers can predict signal variability by combining real-time network telemetry data and traffic forecasting with AI based predictive analytics. This helps in improving the system margin utilisation and reduces cost-per-bit.

AI-defined infrastructure

AI-defined infrastructure (AIDI) can manage planning, building, running and maintenance tasks. In planning tasks, AIDI is used for analysing demand trends and predicting infra requirements in order to plan appropriately. This ensures the infrastructure is according to the requirements.

In building tasks, the necessary resources can be deployed as per the workload requirements. Resources can be de-allocated when there is no need for them. The infrastructure components can be configured easily. In the running and maintenance tasks, AIDI can be used to analyse the data patterns. These patterns help in indicating the behaviour of the system which, in turn, helps in making a model of the system. AI based training helps in building this model with quality parameters like availability, scalability and storage.

The anomalies in the network can be identified by the AIDI based platform. These include intrusion detection, fraud points, fault points, and infrastructure abuse and failure. The platform can detect the threat, and act to rectify and fix the problem. It has features to react or proactively act based on a single or group of infrastructure components. Errors can be fixed completely by autonomous actions. AIDI helps in reducing the cost of IT infrastructure by using the most optimal components.

> AI is reinventing network operations.
>
> —***Bradley Mead,*** *head of network managed services, Ericsson*

Challenges in existing networks

AI can be used to achieve the long desired goal of end-to-end automation, which may remove humans from the equation. Network providers want their networks and operations to become adaptive. This is to respond to a changing, competitive landscape and consumer demands. These demands require a coherent combination of human-controlled and supervised automated operational processes. They might also need analytics-driven intelligence and a programmable infrastructure.

The evolution to 5G and IoT adoption is putting massive pressure on today's networks. There is need to increase the capacity by orders of magnitude. On a related front, networks need to have the ability to respond to unpredictability in traffic patterns. The optical network, which is at the heart of communications, helps to interconnect people, data centres and devices in the network.

The network needs to meet today's demands

Network operators have challenges in handling bandwidth demands today. They are doing so by deploying, managing and sparing different hardware. They use cost-optimised solutions as per specific applications. Operators select the hardware based on the best-guess fibre characterisation data. Lack of network visibility and efficiency is forcing them to operate at sub-optimal capacity. This is leading to loss of revenue, resulting in costly network overbuilds.